

PLATFORM PORTFOLIO & ML ARCHITECTURE SHOWCASE

E-Commerce Price Auditing & Market Segmentation Pipeline

An end-to-end data engineering system built to clean messy, web-scraped grocery data, extract key hidden attributes, and cluster products by structural value tiers.

Data Quality Engineering & Feature Extraction

Integrity Check

1. Automated Price Audit

Web scraping frequently catches flipped values. Our system flag-checks inversion metrics to discard records displaying anomalies where: **Sale_Price > Regular_Price.**

600 × 400

Regex Parsing Engine

2. Structural Feature Extraction

Messy text descriptions are automatically parsed to build normalized unit size and pack count features.

RAW STRING SAMPLE	PACK QTY	UNIT SIZE
Fage Total Greek Yogurt 454g	1	454g
Coca-Cola 12p x 355ml Cans	12	355ml
Chobani Flip 4ct 113g Pots	4	113g

REGEX MATCH SEQUENCE

(\d+)\s*(?:ct|pack|p|x)\s*|\s*(\d+\.\d*)\s*(?:g|ml|oz)

Price Distributions and Savings Landscapes

600 × 400

EDA Findings

Statistical Landscape

- ✓ **Distribution Skew:** Major concentration clusters around the \$1.00 – \$8.00 base category, showing essential consumer pricing.
- ✓ **Premium Outliers:** Specialized multi-packs and family assets form a clean right-tail trend ranging to \$40.00.
- ✓ **Natural Boundaries:** Defined segments validate our decision to construct precise price clustering models.

14.2%

Average Storewide Discount Percentage Detected

Unsupervised Product Tiering via K-Means Clustering

600 × 400

- 0

Budget Essentials

High-volume daily staples

\$0 – \$10
- 1

Mid-Tier Selections

Standard brands & moderate sizing

\$10 – \$20
- 2

Premium Offerings

Specialty goods and bulk imports

\$20+

Promotional Outlier Analysis:
Temporary promotional spikes appear beneath standard baselines. By auditing normal list patterns, our models correctly ignore temporary sales to retain permanent classification accuracy.

Operational Value & Key Takeaways



BUSINESS PHASE

Strategic Business Impact

- ✓ **Competitor Intelligence:** Structuring competitor databases in real-time to align our catalogue positions against rivals.
- ✓ **Margin Maximization:** Instantly detect erroneous promotional pricing inversions to mitigate leakage.
- ✓ **Dynamic Market Mapping:** Categorize newly added competitor items instantly into high, mid, or budget cohorts.



ENGINEERING PHASE

Future Technical Scope

- ✓ **Predictive Elasticity:** Train pricing elasticity curves based on validated historical segment bounds.
- ✓ **Real-time Scraping Ingestion:** Refactor parsing elements to work natively with live-streaming pipeline systems.
- ✓ **Deep Learning Alignment:** Implement transformer architectures to align products with matching specifications across retailers.